Tire/Wheel/Trim Mass Measurement Test

1 Scope

Note: Nothing in this standard supercedes applicable laws and regulations.

Note: In the event of conflict between the English and domestic language, the English language shall take precedence.

1.1 Purpose. To determine the mass of an un-mounted tire, wheel, trim or other component.

1.2 Foreword. The mass of an un-mounted tire, wheel, trim or other component, is measured without stickers, tags or other extraneous objects.

1.3 Applicability. Passenger car, compact spare, and light truck tires.

2 References

Note: Only the latest approved standards are applicable unless otherwise specified.

2.1 External Standards/Specifications.

None

2.2 GM Standards/Specifications.

GMW16590

2.3 Additional References.

National Institute of Standards and Technology certified weights.

3 Resources

3.1 Facilities. Not applicable.

3.2 Equipment.

3.2.1 Scale to measure object mass. If using mechanical scale, minimum increments must be no greater than 50 g. Measurement should be accurate to 1% using National Institute of Standards and Technology (NIST) certified weights.

3.2.2 Use reference weights between 5 and 20 kg that are traceable to the NIST to check the scale before the first test every week. If the reading differs from the expected value by more than 0.05 kg, a calibration should be requested and performed. The scale must read zero before putting on the reference weight.

3.3 Test Vehicle/Test Piece. The test piece can be; tire, wheel, trim, or other component.

3.4 Test Time.

Calendar time:	<1 day
Test hours:	0.1 hour
Coordination hours:	<1 hour

3.5 Test Required Information. Not applicable.

3.6 Personnel/Skills. Not applicable.

4 Procedure

4.1 Preparation. Remove all tags, stickers, labels and other foreign objects from the surface and from inside the tire.

4.2 Conditions.

4.2.1 Environmental Conditions. Not applicable.

4.2.2 Test Conditions. Deviations from the requirements of this standard shall have been agreed upon. Such requirements shall be specified on component drawings, test certificates, reports, etc.

4.3 Instructions.

4.3.1 With the scale platform clear, check to see that the scale reading is zero; if not, zero the scale.

4.3.2 Place object on the scale platform and record the mass (kilograms).

5 Data

5.1 Calculations. Not applicable.

5.2 Interpretation of Results. Not applicable.

5.3 Test Documentation. The test results documentation should include the following:

- Test results report
- Test result data files (XML format)

A sample report is given in Appendix A.

Note: For data being loaded into the Global Tire-Wheel Test System (GTWTS), submit electronically the test results data files (XML format) and other supporting files listed above. GMW16590 defines the electronic file details, (e.g., file types, file naming convention, and detailed file formats) for the GTWTS.

6 Safety

This standard may involve hazardous materials, operations, and equipment. This standard does not propose to address all the safety problems associated with its use. It is the responsibility of the user of this standard to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use.

7 Notes

7.1 Glossary.

Mass: The mass of an object (tire, wheel, trim, or component), after removal of all stickers, tags or other extraneous objects.

7.2 Acronyms, Abbreviations, and Symbols.

GTWTS Global Tire-Wheel Test System

NIST National Institute of Standards and Technology

XML Extensible Markup Language

8 Coding System

This standard shall be referenced in other documents, drawings, etc., as follows:

Test to GMW14998

9 Release and Revisions

This standard was originated in January 2006**.** It was first approved by the Tire Global Subsystem Leadership team in July 2006. It was first published in December 2006.

Issue	Publication Date	Description (Organization)
1	DEC 2006	Initial publication.
2	NOV 2011	Five year refresh. (Tire Global Subsystem Leadership Team)

Appendix A

Figure A1: GTWTS Test Report (Sample Only)

GMW14998 Mass Report

Tire-Wheel Systems

Test Request Number:
PQMS Activity Number:

TEST REQUEST DETAILS

Model Year / Program: Requester:
Test Purpose: Requester's Region:

PART DETAILS

Supplier / Trade Name:	Design Vehicle:	Construction Number:
Load Index / Load (N):	Tire Type:	Tire Size/Load Range:
Speed Rating:	Part Status Code:	Sidewall:
Part Numbers:		Tire Condition:

TEST CONDITIONS

TR Comment: Requested Rim Width (in):

TEST RESULTS

Tire ID	DOT	Tester	Site	TIP App	Mach	Mach ID	Tech	Date (DD-MMM-YYYY)	Mass (kg)
								Average for N :=	
								Standard Deviation:	

Tire ID	Tester	Test Date	Comment	Result Status		
				Approval	Name	Date

FILE ATTACHMENTS

GM Proprietary	Page 1 of 1	Date Printed: